The Labor Market Effects of Immigration Enforcement

Chloe N. East, *University of Colorado Denver and Institute of Labor Economics*

Annie L. Hines, *Cornerstone Research*

Philip Luck, *University of Colorado Denver*

Hani Mansour, *University of Colorado Denver and Institute of Labor Economics*

Andrea Velásquez, *University of Colorado Denver*

We examine the labor market effects of Secure Communities (SC), a police-based immigration enforcement policy implemented in 2008–13. Using variation in implementation across local areas and over time, we find that SC decreased the employment of likely undocumented immigrants. These effects are driven not only by deportations but also by adjustments among immigrants who remain in the United States. Importantly, SC also decreased the employment and hourly wages of US-born individuals. We provide support for two mechanisms that could explain this decline in labor demand: an increase in labor costs that decreases job creation and a reduction in local consumption.

We are grateful to Catalina Amuedo-Dorantes, Francisca Antman, Brian Cadena, Hugh Cassidy, Michael Clemens, Brian Duncan, Brian Kovak, Giovanni Peri, Sebastian Tello-Trillo, and Barton Willage; seminar participants at the University of

Submitted July 30, 2020; Accepted May 24, 2022; Electronically published August 16, 2023.

Journal of Labor Economics, volume 41, number 4, October 2023.
© 2023 The University of Chicago. All rights reserved. Published by The University of Chicago Press in association with The Society of Labor Economists and The National Opinion Research Center. https://doi.org/10.1086/721152

I. Introduction

The share of undocumented immigrants in the US labor force grew rapidly in the 1990s and early 2000s. By 2008, approximately 8 million undocumented immigrants worked in the United States, making up 5.4% of the labor force (Passel and Cohn 2016b). In response, local, state, and federal policies were implemented to reduce undocumented immigration through detentions and deportations. However, how these policies impacted the labor market outcomes of undocumented immigrants and whether they were detrimental or beneficial to US-born workers are still largely unknown (Chassamboulli and Peri 2015).

In this paper, we study how police-based immigration enforcement policies have affected the short-term labor market outcomes of both immigrant and US-born workers. Specifically, we estimate the labor market effects of a federal policy, Secure Communities (SC), which increased information sharing between local law enforcement agencies and the federal government to facilitate the detection and removal of undocumented immigrants. The policy was ultimately adopted by all US counties, and more than 454,000 individuals were removed under SC during our sample period of 2008–14.[1] In addition to the large number of removals, SC may have further reduced the supply of immigrant workers who remain in the United States through chilling effects. This is because fear of interacting with local police or having to present forms of identification likely increased the cost of working outside the home and the cost of job searching (Kohli, Markowitz, and Chavez 2011; Valdivia 2019).[2]

California at Irvine, Syracuse University, Northeastern University, University of Texas at Austin, University of Pittsburgh, and Ohio State University; as well as participants at the National Bureau of Economic Research spring 2021 conference "Immigrants and the US Economy," the Society of Labor Economists annual conference, and the Stanford Institute for Theoretical Economics (SITE) summer workshop on migration for helpful comments and discussions. We are grateful to Reid Taylor, Tyler Collinson, and Evan Generoli for excellent research assistance. We thank Sue Long at Transactional Records Access Clearinghouse (TRAC) for assistance with data on US Immigration and Customs Enforcement deportations, which we obtained from Syracuse University as TRAC fellows. Chloe East was supported by funding from the Office of Research Services at the University of Colorado Denver, and Annie Hines benefited from support from the Russell Sage Foundation, the University of California Mexico Initiative, and the National Institute on Aging (grant T32-AG000186). All errors are our own. Contact the corresponding author, Hani Mansour, at hani.mansour@ucdenver.edu. Information concerning access to the data used in this paper is available as supplemental material online.

[1] Statistics on removals under SC come from Transactional Records Access Clearinghouse (TRAC).

[2] Wang and Kaushal (2018) found that exposure to immigration enforcement policies, including SC, increased mental distress among Latino immigrants. Additionally, Alsan and Yang (2018) found that SC reduced the participation of Hispanic

SC was rolled out on a county-by-county basis between 2008 and 2013 because the Department of Homeland Security was unable to implement it simultaneously across the United States. Our empirical model exploits this variation in timing of SC implementation across local labor markets. This spatial approach estimates the total local labor market effects of SC for likely undocumented immigrants and US-born workers. This is in contrast to models that exploit variation in immigrant flows across education experience groups and instead identify national effects of one group relative to another (Dustmann, Schönberg, and Stuhler 2016).[3] We find no evidence that the timing of SC enactment in a local area can be predicted by pre-SC changes in demographic and economic characteristics, and distributed lag analyses show no significant trends in labor market outcomes before implementation. Thus, the timing of SC implementation can be thought of as plausibly exogenous. Moreover, the relative speed of the rollout, the fact that all US counties eventually adopted SC, and our focus on estimating the short-run effects of the policy limit the scope of within-US mobility by workers or capital, so concerns about spatial arbitrage should be minimal (Borjas 2003; Borjas and Katz 2007; Cadena and Kovak 2016).

We use data from the 2005–14 American Community Survey (ACS) and conduct the analysis at the commuting zone (CZ) level, which provides a measure of local labor markets representing both metropolitan and rural areas (Tolbert and Sizer 1996; Dorn 2009; Autor and Dorn 2013; Autor, Dorn, and Hanson 2013). We merge in information about annual exposure to SC based on the population-weighted share of counties in the CZ that implemented the policy, and we estimate distributed lag and difference-in-difference models, which include CZ and year fixed effects as well as CZ-specific linear trends. The results are robust to adding controls to account for the impacts of the Great Recession and the presence of other immigration enforcement policies. The results are also similar when using the estimator proposed by Callaway and Sant'Anna (2021), which accounts for potential bias in two-way fixed effect models with a staggered rollout design.

We identify the effects of SC on likely undocumented immigrants by focusing on a sample of foreign-born individuals aged 20–64 with a high school degree or less. This sample of low-educated foreign-born (LEFB) individuals captures a large portion of the undocumented population that would have been directly affected by SC. This is a common approach used in the literature to proxy for documentation status, given that it is not asked in large-scale US Census Bureau surveys, including the ACS (Van Hook

citizens in safety net programs for which they were eligible and attributed this to chilling effects.

[3] Dustmann, Frattini, and Preston (2012) show that immigrants downgrade their skills upon arrival, so the education experience group approach can lead to misclassification and can bias the estimated effects of immigration on the labor market (Dustmann, Schönberg, and Stuhler 2016).

and Bachmeier 2013; Passel and Cohn 2014; Borjas and Cassidy 2019; Albert 2021).[4] We also study the effects of SC on US-born individuals aged 20–64. For both demographic groups, we focus on two outcome variables: (1) the employment share of the time-fixed population, which we calculate by dividing the number of employed individuals in the demographic group, CZ, and year by the total working-age CZ population in 2005 (hereafter, "employment share"); and (2) the hourly wage, which we calculate by dividing annual income by annual hours worked for full-time workers in the demographic group.

Focusing first on likely undocumented immigrants, we find that SC reduced their employment share by 5.7%. This decline is partly driven by a decrease in their share in the local population (by 1.2%) and partly by a decrease in employment among those who remain in the United States (by 1.3%). The effect of SC on the wages of LEFB individuals is also negative (1.4%) but marginally insignificant. These negative effects among undocumented immigrants who were not deported is consistent with both the presence of chilling effects and a decline in the demand for their labor because of their higher risk of deportations (Albert 2021).

Turning to the effects of SC on US-born individuals, we find no evidence that, on average, the policy increased their employment share or hourly wages. Instead, the results indicate that SC decreased both the employment share and wages of US-born individuals by about 0.5% and 0.6%, respectively. These changes are not driven by population changes among US-born individuals. Taken together, the results imply that a 1% decrease in the employment share of LEFB individuals is associated with about a 0.1% decline in the employment share and hourly wages of US-born workers. This is consistent with the impacts of historical removal policies that also led to little improvements in the labor market outcomes of native workers (Clemens, Lewis, and Postel 2018) and, in some cases, to declines in their wages and employment (Lee, Peri, and Yasenov 2022).[5]

Informed by the past theoretical and empirical literature, we propose and investigate two potential mechanisms through which SC could have affected the labor market outcomes of US-born workers. First, the decline in the labor

[4] We use the terms "LEFB" and "likely undocumented" interchangeably. However, we note that LEFB likely also includes some documented immigrants. Our results are similar when using more restrictive samples of likely undocumented immigrants and when conditioning on citizenship status instead of country of birth (Orrenius and Zavodny 2009; Amuedo-Dorantes and Bansak 2012).

[5] Several papers have examined the effects of emigration on the labor market outcomes of nonmigrants who remain in sending countries, (e.g., Mishra 2007; Elsner 2013; Dustmann, Frattini, and Rosso 2015; Anelli et al. 2023). We do not directly compare our findings with these papers because removals of existing immigrants vs. departures of emigrants are likely to affect natives in the sending country through different mechanisms.

supply of undocumented immigrants and the increase in their risk of deportation may affect firms' labor demand for substitute and complement US-born workers. On the basis of the job search models developed by Chassamboulli and Peri (2015) and Albert (2021), which assume that US-born workers have higher reservation wages than undocumented immigrants, we posit that the effect on US-born workers who are substitutes in production for undocumented immigrants is ambiguous.[6] This is because while substitute workers face lower competition because of the decline in the employment of LEFB individuals, the rise in the expected wage costs also reduces job creation. In contrast, for workers who act as complements in production to undocumented workers, the decline in job creation is expected to unambiguously decrease labor demand. To examine this empirically, we estimate the heterogeneous effects of SC on US-born workers by education and across different occupational skill categories. The results indicate that while the employment share of low-educated US-born workers increases at the very bottom of the occupational skill distribution, it becomes muted and eventually starts to decrease among workers in the middle and upper parts of the occupational skill distribution. This is consistent with the job competition effect being more salient in lowest-skilled occupations, where the share of LEFB individuals is highest, and with the job creation effect playing an important role for workers in higher-skilled occupations.

The second potential mechanism is a reduction in the demand for local goods, which is also expected to reduce labor demand. The impact of immigration-induced consumption on the labor market has received less attention in the existing literature (Dustmann, Schönberg, and Stuhler 2017).[7] To examine this possibility, we estimate the effects of SC separately for US-born workers in tradable and nontradable industries. Consumption effects should have a larger impact on workers in the nontradable sector (e.g., food services), where demand is local and where prices respond to local shocks (Burstein et al. 2020). Indeed, the negative employment effects of SC on US-born workers are larger in the nontradable sector. We interpret the results as evidence that reduced local consumption may have also played a role

[6] Research shows that undocumented immigrants have lower reservation wages compared with native-born workers with similar skills (Rivera-Batiz 1999; Kossoudji and Cobb-Clark 2002; Pan 2012; Albert 2021).

[7] For example, Greenwood and Hunt (1984) and Altonji and Card (1991) discuss the possibility that immigration inflows can impact the labor market outcomes of native workers through an increase in aggregate demand. Hercowitz and Yashiv (2002) and Bodvarsson, Lewer, and Van den Berg (2008) conclude that the increase in local demand due to the arrival of immigrants can mask a negative labor market impact due to increased competition. Dustmann, Schönberg, and Stuhler (2017) argued that the negative employment effect of Czech migrants on native German workers may be explained by the commuting nature of the migrants and thus a lack of a consumption channel.

in explaining how SC affected the labor market. Our reduced-form approach, however, does not enable us to distinguish the relative importance of each mechanism.

An important contribution of this paper is that it is the first to estimate the short-run labor market effects of a mandatory federal police-based enforcement policy. Prior work has mostly focused on estimating the effects of optional 287(g) agreements, which authorized local law enforcement to act as immigration agents, including checking immigration status among people who were arrested. While these agreements are similar in design to SC, they were voluntarily adopted by only about 50 counties, raising concerns about endogenous policy adoption and making identification more challenging (Pham and Van 2010; Bohn and Santillano 2017). In addition, unlike the prior 287(g) literature, we separately identify the effects of SC on immigrants and natives and examine the role of different mechanisms. The only other paper on the labor market effects of SC focuses on highly educated US-born mothers with young children, who are likely affected through an increase in the cost of outsourcing household production (East and Velásquez 2022). In contrast, we take a broad approach and look at the overall effects of SC on the labor market outcomes of both immigrants and natives in the local area. We also contribute to a small set of papers studying other impacts of SC, including participation in safety net programs among Hispanic citizens (Alsan and Yang 2018), immigrants' health outcomes (Wang and Kaushal 2018), immigrants' marriage patterns (Bansak and Pearlman 2021), and local crime outcomes (Miles and Cox 2014; Hines and Peri 2019). Given the volatility of immigration enforcement policies across the past three presidential administrations, understanding their effects is crucially important for policy makers.

We also contribute to the broader immigration literature by providing contemporary evidence about the impact of immigration policy pull factors (which increase voluntary departures or forced removals) on labor market outcomes rather than immigration push factors (which increase arrivals), which has been the focus of much of the prior research (Clemens, Lewis, and Postel 2018). The effects may not be symmetric because removing immigrants who are already integrated into the labor market is likely different than adding new workers to the labor market, for example, because of different degrees of substitution between these groups of immigrants and natives (Lee, Peri, and Yasenov 2022). This is a key contribution because much of the evidence on the effects of removing immigrants in the United States comes from historical policies (Clemens, Lewis, and Postel 2018; Lee, Peri, and Yasenov 2022), and there have been major shifts in the US immigration system and the economy since then. Additionally, we document that removal policies do not impact labor markets outcomes only through deportations but also through their impact on immigrants who remain in the United States.

II. Policy Background

A. The Secure Communities Program

SC is one of the largest interior immigration enforcement programs administered by US Immigration and Customs Enforcement (ICE).[8] SC facilitated information sharing between local and state law enforcement agencies, the Federal Bureau of Investigation (FBI), and the Department of Homeland Security. Usually, local law enforcement agencies conduct a criminal background check after a person is arrested by sending their fingerprints to the FBI. Prior to SC implementation, fingerprints received by the FBI were not used to check the legal status of a person or their eligibility for removal. Instead, violators of immigration law were identified via interviews conducted by federal agents under the Criminal Alien Program or by local agents authorized to act as immigration agents under 287(g) agreements. Under SC, fingerprints were automatically sent to ICE, which subsequently ran the fingerprints against their biometric database, known as the Automated Biometric Identification System (IDENT), to determine an individual's immigration status.

If the fingerprints were matched, "detainers could be issued when an immigration officer had reason to believe the individual was removable," which could be for criminal reasons or for immigration crime–related reasons.[9] A detainer (or deportation) did not have to be preceded by a conviction. The detainer required state or local law enforcement agencies to hold an arrested individual for up to 48 hours until ICE could obtain custody and start the deportation process. Thus, a detainer prevented the release of individuals whose cases were dismissed and, for those who were charged with a crime, did not provide them the opportunity for a pretrial release through bail. As a result, conditional on an individual being arrested, SC substantially increased the probability of apprehension and deportation of undocumented immigrants by ICE.

When estimating the effects of SC on the labor outcomes of both immigrants and natives, it is important to note that this is a police-based enforcement policy. This is in contrast to employment-based enforcement policies, such as E-Verify, which target undocumented immigrants at their place of employment. However, the operation of SC is likely to have affected immigrant labor supply not only through deportations of people interacting with the criminal justice system but also through chilling effects among those who were not deported.[10] As we show in table 1, among the roughly

[8] For excellent reviews of the SC program, see Cox and Miles (2013), Miles and Cox (2014), and Alsan and Yang (2018). The information in this section comes primarily from these reviews.

[9] https://www.ice.gov/pep.

[10] It is also possible that SC could have changed the number of undocumented immigrants by increasing voluntary out-migration from the United States or by reducing in-migration to the United States.

Table 1
Most Serious Criminal Conviction and Basic Demographic Characteristics of Deportees under SC, 2008–14

Most Serious Criminal Conviction	Share of All Deportees (%)
None	20.63
All violent	18.54
All nonviolent	60.83
DUI	10.94
Traffic	7.01
Immigration	5.46
Marijuana	2.38
Male	95.61
Latin American	92.22

NOTE.—Information on deportees is drawn from the individual listings of all deportations under SC from TRAC, described in the appendix. The most serious criminal conviction may be—but does not have to be—the crime for which the deportee was initially apprehended.

454,000 individuals removed, 21% were not convicted of a crime, and 61% had a nonviolent crime as the most serious offense, indicating that SC increased the risk of deportations of nonviolent and law-abiding undocumented immigrants (Alsan and Yang 2018; Amuedo-Dorantes, Puttitanun, and Martinez-Donate 2019; East and Velásquez 2022).[11] In fact, qualitative evidence suggests that immigrant communities believed that SC allowed police officers to act as ICE agents, and advocacy groups suggested that SC provided a way for law enforcement to use minor violations to target the Hispanic population (Kohli, Markowitz, and Chavez 2011).[12] Thus, in addition to deportations, SC is likely to have increased the cost of interacting with local police, commuting to work, and searching for a job for a broad swath of the immigrant community.

The broad reach of SC and the disruption it caused to immigrant communities led several local jurisdictions, known as sanctuary cities, to refuse to cooperate with ICE detainer requests, arguing that they were unconstitutional under the Fourth Amendment (Alsan and Yang 2018). At the end of 2014, the SC program was replaced by the Priority Enforcement Program (PEP). Under PEP, the same screening process occurred as did under SC,

[11] We use restricted-access data on deportations and detentions under SC from TRAC at Syracuse University to provide context for understanding the potential effects of SC. Details about this data can be found in the appendix (available online). To classify violent crimes, we followed the definitions of the FBI's Uniform Crime Reporting Program (https://ucr.fbi.gov/crime-in-the-u.s/2010/crime-in-the-u.s.-2010/violent-crime). The top violent crimes leading to a deportation in this period were assault, burglary, domestic violence, and robbery.

[12] Immigrants from Latin American countries were overrepresented among those deported under SC; in 2007, undocumented immigrants from Latin American countries accounted for 70% of the undocumented population of immigrants in the United States (Passel and Cohn 2011), but table 1 shows that 92% of those deported were from a Latin American country.

but PEP focused more on individuals convicted of serious crimes or those who were deemed to pose a threat to public safety.[13]

B. Implementation of Secure Communities

Unlike previous police-based enforcement programs, SC is a federal program, and local and state law agencies could not opt in or opt out of SC. For empirical purposes, this is important because local agencies had much more limited discretion in the usage of the program compared with other interior immigration enforcement polices, such as 287(g) agreements (Miles and Cox 2014). Moreover, despite being a federal program, because of resource and technological constraints, SC was rolled out on a county-by-county basis between 2008 and 2013, until the entire country was covered (Miles and Cox 2014; Alsan and Yang 2018).

Our identification strategy, described in more detail below, relies on the piecemeal implementation of SC across counties. Therefore, it is important that the timing of the rollout across counties not be related to time-varying county characteristics. Cox and Miles (2013) show that the choice of the earliest activations of SC were related to the fraction of the county's Hispanic population, distance from the US-Mexico border, and presence of local 287(g) agreements. Importantly, however, they show that early adopters were not selected in terms of the county's economic performance, crime rates, and potential political support for SC. They also explain that while early adoptions were done on a county-by-county basis and were therefore more targeted, the timing of adoption as the rollout progressed was more random since it became common for multiple inactive counties within the same state to be simultaneously activated on the same date (mass activations). This pattern can be seen in figure 1, which plots the rollout of SC across counties and over time. In our main sample, we include the whole country, but the results are robust to excluding early-adopter areas.

We gathered information on the rollout dates of SC from ICE, and we also examine whether changes in pre-SC demographic and economic characteristics between 2005 and 2007 at the CZ-level predict the year when SC was adopted. Columns 1 and 2 of table 2 report the average and standard deviation of changes in CZ characteristics, respectively. In column 3, we report estimates of the relationship between changes in CZ characteristics from 2005 to 2007 and the year of SC adoption in the CZ. The dependent variable is the calendar year of CZ adoption, and we include all changes in characteristics together in one regression. These characteristics include the CZ immigrant population, local 287(g) agreements, and Bartik-style measures of labor demand. We construct four Bartik-style measures of labor demand that correspond to the following four demographic groups: (1) US-born working-age adults, (2) foreign-born working-age adults, (3) working-age

[13] SC was reactivated in 2017 and suspended again in 2021.

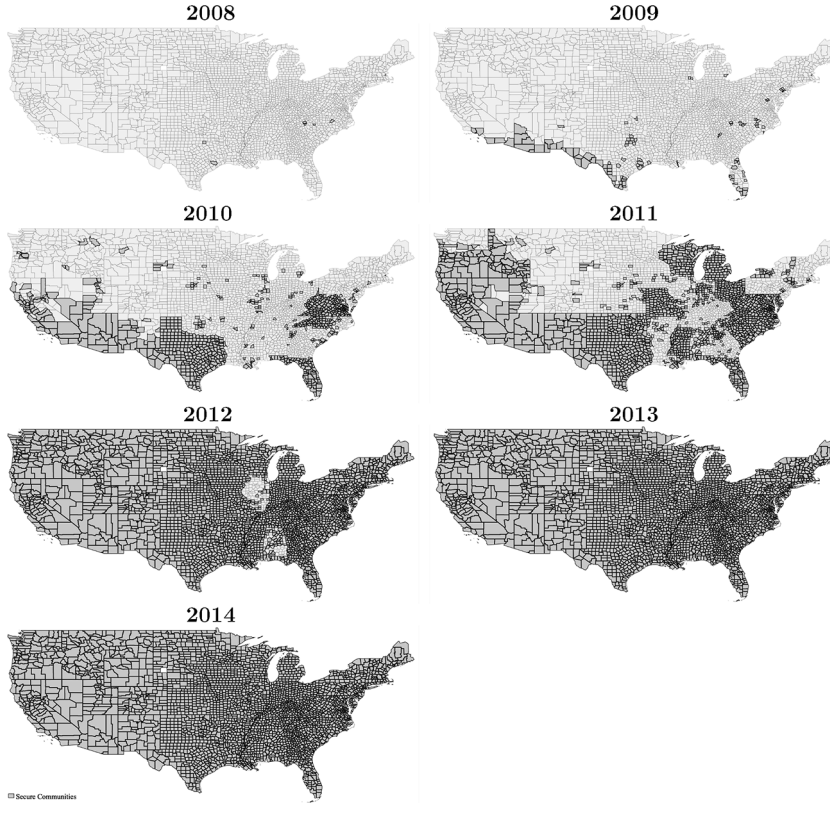


FIG. 1.—Rollout of SC across counties, by year. Shading indicates counties that adopted SC during the calendar year. Rollout dates were collected from the Department of Homeland Security.

adults with more than a high school diploma, and (4) working-age adults with a high school diploma or less. For each group, we calculate the group-specific CZ-level employment by industry as a fraction of total group-specific CZ employment in 2005. We then multiply these industry shares by the industry-specific national (not group-specific) employment for all working-age adults in each year. This allows us to have a measure of predicted labor demand for each group by CZ, on the basis of the industries that they work in during the base period and the overall changes in employment in these industries over time. See the appendix for additional details on the CZ-year level control variables.

Out of 10 pre-SC characteristics, there are only two statistically significant relationships: the change in 287(g) jail agreements and the change in the male noncitizen population. However, although significant, the magnitude for 287(g) agreements is small: an increase of 1 standard deviation in

Table 2
Correlation between CZ Adoption Year of SC and Changes in CZ Characteristics in 2005–7

	Characteristic		
	Mean (1)	SD (2)	Regression Estimate (3)
Change in % noncitizen	.06	.21	.357
			(.271)
Change in % male noncitizen	.09	.32	.370*
			(.215)
Change in % low-educated male noncitizen	.13	.57	.066
			(.116)
Change in % Hispanic low-educated male noncitizen	.27	1.90	.022
			(.026)
Change in task 287(g)	.01	.09	.342
			(.660)
Change in jail 287(g)	.04	.14	−2.188***
			(.516)
Change in US-born individuals, Bartik	49.49	78.48	−.027
			(.031)
Change in foreign-born individuals, Bartik	44.01	70.57	.003
			(.020)
Change in low-educated individuals, Bartik	44.39	69.90	−.009
			(.031)
Change in high-educated individuals, Bartik	49.55	77.87	.032
			(.030)
Mean Y			2,010.10
R^2			.13
N			740

NOTE.—Data on CZ characteristics are drawn from the 2005–7 ACS. See sec. II and app. sec. A.1 for more detailed descriptions of these variables. We estimate the following regression: $\text{year}_j = \alpha + \theta \Delta W_j' + \epsilon_j$, where year_j is the year SC was implemented in the CZ_j. $\Delta W_j'$ includes changes in CZ-level demographics, economic conditions, and 287(g) agreements between 2005 and 2007. The regressions are weighted by the CZ population in 2000.

* $p < .10$.

*** $p < .01$.

exposure to 287(g) jail agreements is associated with a 3.7-month earlier adoption of SC ($-2.188 \times 0.14 \times 12$). For the male noncitizen population, the magnitude is also small: an increase of 1 standard deviation is associated with a 1.4-month later adoption of SC ($0.370 \times 0.32 \times 12$). This further supports the idea that the timing of the rollout can be thought of as good as random.

III. Data

A. Employment and Wage Outcomes

To construct the analysis sample, we merge information on the rollout dates of SC with local-level employment and wage data drawn from the 2005–14 ACS (Ruggles et al. 2017). The ACS is a repeated cross-sectional

dataset covering a 1% random sample of the United States, and it is particularly well suited for our purposes because it is nationally representative and includes local labor market identifiers across the United States. In addition, it includes a rich set of demographic characteristics, including country of birth, enabling us to calculate labor market outcomes for different groups of foreign-born and US-born individuals.

We begin our sample in 2005, as this is the first year the detailed geographic identifiers used to define the CZs are in the public-use data. Our sample ends in 2014 when SC was replaced by PEP. We chose not to extend the analysis to later years in order to exclusively focus on the effects of SC. We conduct our analysis at the CZ level, but the ACS data identify only Public Use Microdata Areas (PUMAs), so we concord PUMAs to CZs using 1990 CZ definitions and the crosswalks in Dorn (2009) and Autor and Dorn (2013). Similarly, we concord the county-level SC data to the CZ level by weighting the SC value for each county by the county's population share in the CZ, similar to the approach taken by Watson (2013) and Alsan and Yang (2018). In addition, since the ACS data include only the year in which the survey was conducted, we create a variable that indicates the fraction of the survey year SC was in place in each CZ. The main advantage of using CZs as our unit of analysis is that they are designed to provide a measure of local labor markets while representing both metropolitan and rural areas (Dorn 2009), but our results are robust to conducting our estimation at the PUMA level rather than at the CZ level.

To estimate the effects of SC on employment, we construct the employment shares of the time-fixed CZ population for likely undocumented and US-born workers by dividing the number of employed individuals in each group at the CZ-by-year level by the total working-age CZ population in 2005. We multiply these shares by 100 to ease the presentation: $(\text{Emp}_{jt}/\text{Pop}_{j2005}) \times 100$, where j indexes CZ and t indexes survey year. To calculate both the numerator and the denominator, we use the ACS-provided person-level weights. Our primary samples are working-age (ages 20–64) LEFB individuals (high school degree or less) and working-age US-born individuals.[14] In both samples, we exclude workers in the military and public administration sectors

[14] Our primary definition of likely undocumented immigrants does not condition on citizenship status because of concerns about misreporting by undocumented immigrants (Van Hook and Bachmeier 2013; Brown et al. 2018). One additional concern with our definition of likely undocumented is that some undocumented immigrants might choose not to participate in surveys conducted by the US government (Passel and Cohn 2011; Hoefer, Rytina, and Baker 2012; Warren and Warren 2013; Van Hook et al. 2014; Genoni et al. 2017; Brown et al. 2018). While previous studies estimate an overall 7.5% undercount of undocumented immigrants (Warren 2014), we are unable to assess how the undercount varies in response to SC. This undercount is likely to bias the estimates of the LEFB population toward zero, while it should not impact the estimated effects on the US-born population.

because of the potential direct impact of SC on their employment, but our results are robust to including these sectors. Given the construction of this employment share variable, SC will affect only the numerator. We use this as our primary measure of employment because it captures employment changes of likely undocumented immigrants through both potential mechanisms: (1) changes in the population of workers through both voluntary and involuntary migration and (2) changes in employment of those remaining in the United States. We do not focus on the employment-to-population ratio (or employment rate) used in other studies on the labor market impacts of immigration (e.g., Altonji and Card 1991; Dustmann, Fabbri, and Preston 2005; Boustan, Fishback, and Kantor 2010) because it captures only the effects of SC on the employment of immigrants who remain in the United States and is thus likely to understate the total employment effects of SC on likely undocumented immigrants (Dustmann, Schönberg, and Stuhler 2017). However, to shed light on the role of these two mechanisms, we also estimate the effects of SC on (1) the population share of time-fixed CZ population and (2) the employment-to-population ratio for both LEFB and US-born individuals. We use the employment share as our main measure of employment for US-born individuals to easily compare the results across both groups (though we show that the effects on the employment-to-population ratio of US-born workers are very similar).

We complement our employment analysis by estimating the effect of SC on hourly wages. Hourly wages are not directly reported in the ACS, so we calculate them by dividing annual earnings by annual hours.[15] We then use the ACS-provided person-level weights to calculate average group-specific wages for each CZ and year. As is common in the literature, we use a sample of full-time LEFB and US-born workers aged 20–64 who report to usually work 30 or more hours per week (Dustmann, Schönberg, and Stuhler 2017). All wages are in constant 2014 dollars.

Since our sample period spans the Great Recession, we test the robustness of the results to accounting for CZ time-varying economic conditions that may influence employment by including Bartik-style measures of labor demand (Bartik 1992) for US-born workers and foreign-born workers separately (described in sec. II.B and the appendix), measures of the housing boom and bust, and region × year fixed effects. We also check the robustness of the results to controlling for the presence of 287(g) agreements, E-Verify, and sanctuary city policies.[16]

[15] We calculate annual hours by multiplying usual hours of work per week by weeks worked last year. Because weeks of work are reported in intervals in 2006–14, we use the midpoint of the reported interval as the value for weeks of work. Specifically, weeks values of 8, 20.8, 33.1, 42.4, 48.3, and 51.9 are used for the reported intervals 1–13, 14–26, 27–39, 40–47, 48–49, and 50–52, respectively. To adjust for outliers, observations are winsorized at the 2.5 and 97.5 percentile wage value.

[16] Details on these programs and variable construction are in the appendix.

We provide summary statistics for all the main variables in table A1 (tables A1–A5 are available online). The average employment share of LEFB individuals across CZs is 6.75%. The average employment share of US-born workers is 58.61%. The total employment share for all individuals of working-age is 71.39% (not shown in the table), so LEFB individuals make up roughly 9.5% of all workers in this time period. Turning to wages, we find that the average wage of full-time LEFB individuals is $15.60 per hour, and the average wage of US-born full-time workers is $24.39.

IV. Empirical Strategy

The empirical strategy uses the geographic and temporal variation in the implementation of the SC program to identify its effects on the employment and wages of likely undocumented and US-born workers. Specifically, we estimate the following model:

$$Y_{jt} = \alpha + \beta \mathrm{SC}_{jt} + X'_{jt}\gamma + \nu_j + \lambda_t + t\delta_j + \epsilon_{jt}, \tag{1}$$

where SC_{jt} is a continuous variable indicating CZ-level exposure to SC and ranges between 0 and 1. Once SC has been implemented by January of year t in all counties in a CZ j, the variable SC_{jt} takes a value of 1 for the remainder of the sample. Therefore, β measures the effect of 100% of the CZ population being covered by SC for the entire survey year. The model includes year fixed effects, λ_t, to account for national economic shocks, and CZ fixed effects, ν_j, to control for CZ time-invariant unobserved heterogeneity, such as proximity to the border. To account for differential trends in labor outcomes across areas, we include CZ-specific linear trends, $t\delta_j$.[17] Our fully specified model also includes a vector of Bartik-style measures of labor demand, X_{jt}, which controls for time-varying CZ-specific economic conditions. The model is weighted by the CZ population in 2000, which ensures that we are estimating the policy-relevant effect of SC on the average individual. We cluster the standard errors by CZs.

The underlying identification assumption is that there were no time-varying CZ-specific factors correlated with the timing of the adoption of SC. To provide support for this assumption, we follow Suarez, Serrato, and Zidar (2016), Fuest, Peichl, and Siegloch (2018), and Schmidheiny and Siegloch (2020) and estimate the following distributed lag model:

$$Y_{jt} = \alpha + \sum_{k=-2}^{2} \gamma_k \mathrm{SC}_{j,t-k} + X'_{jt}\gamma + \nu_j + \lambda_t + t\delta_j + \epsilon_{jt}, \tag{2}$$

where the estimated γ_k's measure the relationship between SC and the change in the labor market outcome of interest. The pretreatment effect

[17] The results are similar if we instead model only pretrends and use this to predict posttreatment trends (Wolfers 2006; Lee and Solon 2011; Goodman-Bacon 2021b; Borusyak, Jaravel, and Spiess 2022).

changes are denoted by the leads ($k < 0$), and the effects of SC after adoption on changes in outcomes are denoted by the lags ($k \geq 0$). The distributed lag model allows us to use a continuous treatment variable, so it is more comparable to our baseline model. However, to calculate treatment effects that represent the total effect of SC before and after implementation, we need to choose a reference period and then sum the distributed lag coefficients, moving away from this reference period. This is because the γ_ks measure only changes in the treatment effect. We denote the total effects that we plot as β_ks. We follow the convention used in event study models and choose the period $k = -1$ as our reference period, so we set $\beta_{-1} = 0$. For periods $k < -1$, we cumulate negatively so that $\beta_{-2} = -\gamma_{-1}$, $\beta_{-3} = -\gamma_{-1} - \gamma_{-2}$, and so on. For periods $k > -1$, we cumulate positively so that $\beta_0 = \gamma_0$, $\beta_1 = \gamma_0 + \gamma_1$, and so on (Schmidheiny and Siegloch 2020). If the parallel trends assumption holds, the estimates of β_k where $k < 0$ should be close to zero and statistically insignificant.

V. Results

A. Effects on Employment and Wages

We begin by estimating the labor market effects of SC on likely undocumented immigrants. Estimates of equation (1), where we control for CZ and year fixed effects as well as CZ-specific linear trends, are presented in panel A of table 3. The results in column 1 indicate that SC reduced the employment share of LEFB individuals by 0.398 percentage points (significant at the 1% level). We add controls for Bartik-style measures of labor demand in panel B, which capture CZ-level differences in the impact of the Great Recession. Reassuringly, these controls do not impact the estimated effects, suggesting that the timing of SC implementation was not correlated with predicted changes in local economic conditions. Including these Bartik-style controls is our preferred specification, and with this, we find that SC reduced the employment share of LEFB individuals by about 5.74% relative to their mean (−0.387/6.75). Although our identification strategy is not designed to estimate nationwide effects, we compare the decline in the employment share of LEFB individuals to the total number of deportations as a share of the 2005 US population. If we assume that deportations were evenly distributed across CZs, the share of deportees in each CZ would be equal to 0.262 ($454{,}000/(1.73 \times 10^8)$). Comparing this with our estimate of 0.387 suggests that mechanisms other than deportations may be important.

The effects of SC on the log hourly wage of likely undocumented immigrants are presented in column 4 of table 3. As with the effects on employment, the estimates do not change when we control for local impacts of the Great Recession in panel B. We find that SC reduced the average hourly wage of LEFB individuals by 1.4%, and this estimate is marginally insignificant ($p = .13$). The wage results are estimated with less precision than the employment results, which may be due to the potential measurement error

Table 3
Effects of SC on Employment Share and Log Hourly Wages of LEFB Individuals

	(Total Group Employment/ Total CZ Population in 2005) × 100			Log Hourly Wages		
	All (1)	Females (2)	Males (3)	All (4)	Females (5)	Males (6)
	A. Baseline					
β: SC	−.398***	−.0062	−.336***	−.013	−.007	−.016
	(.113)	(.041)	(.085)	(.009)	(.015)	(.012)
CZ-year trends	Yes	Yes	Yes	Yes	Yes	Yes
Bartiks						
Y mean	6.75	2.55	4.21	2.74	2.59	2.81
% effect	−5.90	−2.43	−8.00	. . .	. . .	. . .
Observations	7,400	7,400	7,400	7,383	7,331	7,367
	B. Add Bartiks					
β: SC	−.387***	−.061*	−.326***	−.014	−.006	−.018
	(.093)	(.032)	(.075)	(.009)	(.015)	(.012)
CZ-year trends	Yes	Yes	Yes	Yes	Yes	Yes
Bartiks	Yes	Yes	Yes	Yes	Yes	Yes
Y mean	6.75	2.55	4.21	2.74	2.59	2.81
% effect	−5.74	−2.41	−7.75	. . .	. . .	. . .
Observations	7,400	7,400	7,400	7,383	7,331	7,367

NOTE.—Data are drawn from the 2005–14 ACS. The sample of LEFB individuals includes working-age (ages 20–64) foreign-born individuals with a high school degree or less. Columns 2, 3, 5, and 6 split the sample by sex. In cols. 1–3, the employment share is measured as total employment in the demographic group by CZ and year, divided by total working-age CZ population in 2005, and then multiplied by 100. In cols. 4–6, hourly wages are calculated by dividing annual earnings by annual hours worked among full-time workers. See sec. III for more details. All models include CZ and year fixed effects and CZ-specific linear trends. Panel B also includes Bartik-style controls for local labor demand. The results are weighted using the CZ population in 2000. Standard errors (in parentheses) are clustered by CZ.

* $p < .10$.

*** $p < .01$.

in the wage variable. It is important to note that wages are observed for only currently employed workers. Although we cannot assess directly how the employment and wage outcomes of deportees compare with the average LEFB individual, we find little evidence that SC changed the composition of LEFB individuals who were not deported (in terms of age, ethnicity, number of years in the United States, and educational attainment). Nonetheless, if we assume that deported LEFB individuals had lower wages and were less likely to be employed than the average LEFB individual, then we would conclude that the labor market effects of an enforcement policy that targets the average LEFB individual would be larger (in absolute value) compared with a police-based enforcement policy.

Since most deportees under SC were male, we expect the policy to have had a larger impact on the employment share of males compared with females. The results in columns 2 and 3 of table 3 show that while SC reduced the employment share of LEFB males by 0.326 percentage points, it decreased the

employment share of LEFB females by only 0.061 percentage points (significant at the 1% and 10% levels, respectively). In relative terms, the results imply that the effect of SC on the employment share of LEFB males is three times as large as the effect for LEFB females (7.75% vs. 2.41%). Although statistically insignificant, the effect on hourly wages is also larger for male immigrant workers (col. 5) relative to female immigrant workers (col. 6). Overall, these results show that although the labor market effects of SC are stronger for males, we cannot rule out that SC had a negative impact on the labor market outcomes of LEFB females.

We next estimate the effect on the employment share and hourly wages of US-born workers, and these results are presented in table 4. Similar to our findings for LEFB individuals, the results are not sensitive to adding Bartik-style controls. The results provide no evidence that the average employment share or hourly wages of US-born workers increased after the implementation

Table 4
Effects of SC on Employment Share and Log Hourly Wages of US-Born Individuals

	(Total Group Employment/ Total CZ Population in 2005) × 100			Log Hourly Wages		
	All (1)	Females (2)	Males (3)	All (4)	Females (5)	Males (6)
	A. Baseline					
β: SC	−.329**	−.112	−.217*	−.005**	−.006	−.006**
	(.163)	(.091)	(.111)	(.003)	(.003)	(.003)
CZ-year trends	Yes	Yes	Yes	Yes	Yes	Yes
Bartiks						
Y mean	58.61	28.39	30.22	3.18	3.05	3.28
% effect	−.56	−.39	−.72	. . .	. . .	. . .
Observations	7,400	7,400	7,400	7,400	7,400	7,400
	B. Add Bartiks					
β: SC	−.300*	−.083	−.217*	−.006**	−.006*	−.006**
	(.168)	(.084)	(.116)	(.003)	(.003)	(.003)
CZ-year trends	Yes	Yes	Yes	Yes	Yes	Yes
Bartiks	Yes	Yes	Yes	Yes	Yes	Yes
Y mean	58.61	28.39	30.22	3.18	3.05	3.28
% effect	−.51	−.29	−.72	. . .	. . .	. . .
Observations	7,400	7,400	7,400	7,400	7,400	7,400

NOTE.—Data are drawn from the 2005–14 ACS. The US-born sample includes working-age (ages 20–64) individuals. Columns 2, 3, 5, and 6 split the sample by sex. In cols. 1–3, the employment share is measured as total employment in the demographic group by CZ and year, divided by total working-age CZ population in 2005, and then multiplied by 100. In cols. 4–6, hourly wages are calculated by dividing annual earnings by annual hours worked among full-time workers. See sec. III for more details. All models include CZ and year fixed effects and CZ-specific linear trends. Panel B also includes Bartik-style controls for local labor demand. The results are weighted using the CZ population in 2000. Standard errors (in parentheses) are clustered by CZ.

* $p < .10$.
** $p < .05$.
*** $p < .01$.

of SC, contrary to the predictions of the canonical labor supply and demand model under the assumption that LEFB and US-born workers are substitutes. Instead, SC reduced the employment share of US-born workers by 0.3 percentage points (significant at the 10% level), which is a 0.51% decline relative to the mean employment share (col. 1). On the basis of the 95% confidence interval of this estimate, we can rule out employment effects that are smaller than −0.63 percentage points and bigger than 0.03 percentage points, which provides little evidence that US-born workers benefit from increased immigration enforcement. Additionally, SC decreased hourly wages of US-born workers by 0.6% (col. 4; statistically significant at the 5% level). On the basis of the 95% confidence interval of this estimate, we can rule out wage effects that are smaller than −1.2% and bigger than −0.012%. The decline in both the employment share and wages of US-born workers is consistent with SC reducing the demand for their labor.

In contrast to the effects on LEFB individuals, it is a priori unclear whether the effects of SC should differ between male and female US-born workers. The results in columns 2 and 3 of panel B in table 4 indicate that SC reduced the employment share of male US-born workers by 0.217 percentage points, about 0.72% relative to the mean (statistically significant at the 10% level). The effect of SC on the employment share of female US-born workers (col. 2) is about half as large (0.29%) and is not significant at conventional levels. These results could be explained by the higher degree of overlap in the employment of male LEFB and male US-born workers across sectors, shown in figure A1 (figs. A1–A3 are available online). This figure plots the employment share of male and female LEFB and US-born workers across sectors in 2005. Sectors with the highest share of male LEFB workers, such as agriculture and construction, also employ a much larger share of male US-born workers relative to female US-born workers. For wages, we find a less clear difference by sex (cols. 5, 6).

To corroborate the validity of the research design, figures 2*A* and 2*B* plot the dynamic effects of SC, before and after implementation, for LEFB individuals estimated with equation (2). Figures 2*C* and 2*D* plot these dynamic effects for US-born individuals. In all panels, the βk's in the pre-SC periods are small and statistically indistinguishable from zero, consistent with the parallel trends assumption. In each panel, in the postperiod, we include the corresponding difference-in-difference estimates (reported in cols. 1 and 4 of tables 3 and 4) as a solid gray line for ease of comparison. Although we lack precision, the distributed lag estimates provide evidence of a decline in the employment share and wages of LEFB individuals after SC implementation that are similar in magnitude to the difference-in-difference estimates. We also see similar patterns for US-born workers. The loss in precision may be because the distributed lag model is a more demanding specification. Nevertheless, this test confirms that our estimates are not being driven by differential pretrends across CZs.[18]

[18] We also report these β_ks and their standard errors in table A2.

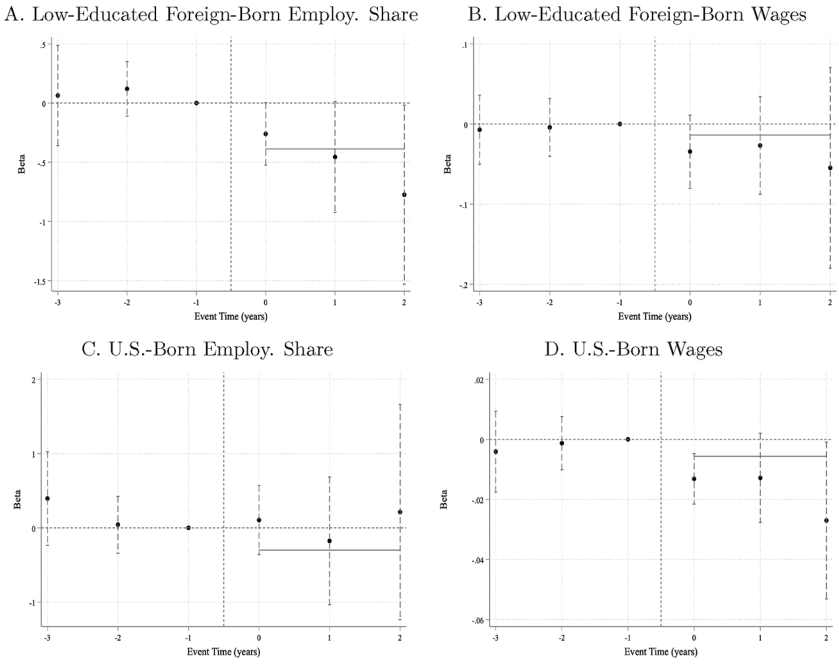


FIG. 2.—Distributed lag models: effects of SC on employment share and wages of LEFB and US-born individuals. Data are drawn from the 2005–14 ACS. The samples are based on working-age (ages 20–64) low-educated (high school or less) foreign-born individuals and US-born individuals. In *A* and *C*, the employment share is measured as total employment in the demographic group by CZ and year, divided by total working-age CZ population in 2005, and then multiplied by 100. In *B* and *D*, hourly wages are calculated by dividing annual earnings by annual hours worked among full-time workers. Distributed lag model γ_k coefficients are estimated using equation (2) and then transformed into total treatment effects (β_ks; circles). Dashed lines indicate 95% confidence intervals of the β_ks. The models include CZ fixed effects, year fixed effects, CZ-specific linear trends, and Bartik-style controls. The solid gray line in the post-SC period corresponds to the difference-in-difference estimates reported in tables 3 and 4. All results are weighted using the CZ population in 2000, and all standard errors are clustered by CZ.

For some outcomes, there is an increase in effect size over time after SC implementation, which could imply that the treatment effects are dynamic, indicating that the difference-in-difference estimates may be biased (Goodman-Bacon 2021a). It is important to note, however, that the estimates could grow over time because SC is phased in across counties within the CZs at different times. To test whether heterogeneous treatment effects across CZs bias our estimates, we implement the estimator proposed by Callaway and Sant'Anna (2021). The advantage of this approach is that it avoids using earlier treated units as controls for later treated units, which are comparisons

that can lead to biased estimates. The results are presented in gray in figure A2, along with the estimates from the distributed lag model in equation (2) in black.[19] The results from the Callaway and Sant'Anna method follow the same general patterns as in the distributed lag model. Overall, the confidence intervals are smaller when using the Callaway and Sant'Anna method, possibly because these methods do not calculate standard errors in the same way. Also, there are a few coefficients that are statistically significant with Callaway and Sant'Anna that are not in the distributed lag model. However, except for two coefficients in figure A2*C*, the 95% confidence intervals overlap across the two methods.

B. Changes in Population versus Employment

Given the way we construct the employment share variable, the decline for LEFB individuals could be due to changes in migration or changes in employment among LEFB individuals who remain in the United States. We first investigate the migration channel by estimating whether SC reduced the population share of LEFB individuals, which—analogous to the employment share—is calculated by dividing the population of LEFB individuals in year t and CZ j by the working-age CZ population in 2005 and multiplying by 100 ($(\text{Pop}_{jt}/\text{Pop}_{j2005}) \times 100$). The result (col. 1 of panel A in table 5) indicates that SC reduced the population share by 0.119 percentage points, or 1.18% relative to the mean ($p = .16$). We also split the sample by sex in columns 2 and 3, and the results are consistent with the fact that most of those deported under SC were male. The male likely undocumented population share declined by a significant 1.87% compared with an insignificant decline of 0.42% for females. If we assume that all deported individuals were employed, these results imply that changes in the population of overall (and male) LEFB individuals can explain up to one-third of the overall (and male) decline in their employment share ($-0.119/-0.387$ for all and $-0.099/-0.326$ for males).[20]

[19] The models in this figure do not control for Bartik-style local labor demand because of differences in the way time-varying controls are included across the two estimators. The Callaway and Sant'Anna method does not allow for a continuous treatment, so we dichotomize the SC variable to equal 1 once a CZ is first treated and 0 in prior years. To account for linear trends with the Callaway and Sant'Anna estimator, we use a two-step method (Goodman-Bacon 2021b). The first step estimates a linear trend for each CZ and outcome using preperiod data only. The second step detrends the data using the extrapolated trend. We use these detrended variables as our outcomes.

[20] To further verify that SC affected the undocumented population, we estimate its effect on detentions using restricted-access data from TRAC, which is described in the appendix. Table A3 reports the impact of SC on the number of detentions at the CZ level scaled by the total working-age CZ population in 2005. With the full set of controls as described in eq. (1), the results in col. 2 indicate that SC increased the detention share of the time-fixed population by 0.118 percentage points, an increase of about 118% relative to the mean detention rate. We are unable to directly

Table 5
Effects of SC on Population Share and Employment-to-Population Ratio of LEFB and US-Born Individuals

	LEFB Individuals			US-Born Individuals
	All (1)	Females (2)	Males (3)	All (4)
	A. Effect on Population			
β: SC	−.119	−.020)	−.099*	−.135
	(.084)	(.043)	(.056)	(.182)
CZ-year trends	Yes	Yes	Yes	Yes
Bartiks	Yes	Yes	Yes	Yes
Y mean	10.13	4.84	5.29	87.52
% effect	−1.18	−.42	−1.87	−.15
Observations	7,400	7,400	7,400	7,400
	B. Effect on Employment-to-Population Ratio			
β: SC	−.885**	.092	−1.812***	−.501***
	(.444)	(.515)	(.628)	(.174)
CZ-year trends	Yes	Yes	Yes	Yes
Bartiks	Yes	Yes	Yes	Yes
Y mean	66.12	52.96	77.73	67.02
% effect	−1.34	.17	−2.33	−.75
Observations	7,397	7,389	7,390	7,400

NOTE.—Data are drawn from the 2005–14 ACS. The sample in cols. 1–3 includes working-age (ages 20–64) foreign-born individuals with a high school degree or less, and this sample is split by sex in cols. 2 and 3. The sample in col. 4 includes all working-age (ages 20–64) US-born individuals. The population share (outcome in panel A) is calculated as the group-specific population in CZ *j* and year *t*, divided by the total working-age CZ population in 2005, and multiplied by 100 ($(\text{Pop}_{jt}/\text{Pop}_{j2005}) \times 100$). The employment-to-population ratio (outcome in panel B) is measured as the group-specific employment in CZ *j* and year *t*, divided by the group-specific CZ population in year *t*, and multiplied by 100 ($(\text{Emp}_{jt}/\text{Pop}_{jt}) \times 100$). The models include CZ fixed effects, year fixed effects, CZ-specific linear trends, and Bartik-style controls. The results are weighted using the CZ population in 2000. Standard errors (in parentheses) are clustered by CZ.

* $p < .10$.
** $p < .05$.
*** $p < .01$.

Next, we estimate the effect on the employment-to-population ratio of LEFB individuals. To construct this variable, we take the same numerator as we used to calculate the employment share, but the denominator is instead the time-varying group-specific population in each CZ *j* and year *t*: $(\text{Emp}_{jt}/\text{Pop}_{jt}) \times 100$ (US Bureau of Labor Statistics 2021). This allows us to examine the employment response among LEFB individuals who remain in the United States. The results indicate that SC reduced the employment-to-population ratio of LEFB individuals by 0.885 percentage points, or 1.34% relative to the mean (col. 1 of panel B in table 5). This decline in the overall employment-to-population ratio is driven by likely undocumented

estimate the impact of SC on deportations because our dataset on deportations does not include data prior to the implementation of SC. Moreover, we chose not to implement an instrumental variables strategy using detention data, since the increase in detentions is only one potential mechanism through which the effects operate.

male workers (col. 3) and is due to an increase in their unemployment rather than changes in their labor force participation (cols. 1 and 3 of table A4). This is consistent with either a chilling effect causing immigrants to take longer to find a job or a decrease in demand for their labor, which we return to in more detail below.

In contrast to the effect of SC on the population share of LEFB individuals, we find little evidence that the policy changed the population share of US-born individuals. Although the point estimate of −0.135 (col. 4 of panel A in table 5) is close in magnitude to the estimate for LEFB individuals, it is not statistically significant and corresponds to only a 0.15% change relative to the mean. This result supports our research design, as it indicates that SC did not impact the migration decisions of US-born individuals. The effects of SC on US-born workers instead operate through a reduction in their employment-to-population ratio. Specifically, we find that SC reduced the employment-to-population ratio of US-born workers by 0.501 percentage points, or 0.75% relative to the mean (significant at the 1% level). The fact that the effects for US-born individuals are similar whether we use a time-fixed (employment share) or time-varying (employment-to-population ratio) denominator (−0.51% compared with −0.75%) and have overlapping 95% confidence intervals provides further evidence that population changes are not an important mechanism for US-born workers. Again, these effects operate through an increase in unemployment, which is consistent with a decline in labor demand for US-born workers (cols. 2 and 4 of table A4).

C. Robustness Checks

We first explore the robustness of the results for LEFB individuals to using alternative definitions of likely undocumented immigrants in table 6, since documentation status for immigrants is not available in the ACS. The results on the employment share and hourly wages are reported in panels A and B, respectively. In column 1, we restrict the sample to include only Hispanic LEFB individuals. In column 2, we restrict the sample to Hispanic LEFB individuals who entered the United States after 1980, and in column 3, we restrict the sample to LEFB individuals who were born in Mexico or Central America and entered the United States after 1980 (Passel and Cohn 2014, 2016a; Warren 2014; Borjas and Cassidy 2019).[21] The results in panel A are similar across these different definitions, indicating that SC reduced the employment share of likely undocumented immigrants by about 8.59%–9.04%. These results are slightly larger but similar to the effect on our main sample of all LEFB individuals (5.74%). Although statistically insignificant,

[21] Previous literature has used 1980 as a cutoff year to classify likely undocumented immigrants because immigrants who arrived in the United States before 1982 could have been legalized under the Immigration Reform and Control Act (Warren 2014).

Table 6
Effects of SC on Employment Share and Log Hourly Wages Using Alternative Measures of Likely Undocumented Immigrants

	Hispanic (1)	Hispanic, Arrived after 1980 (2)	Mexico/Central America, Arrived after 1980 (3)
	A. Dependent Variable: (Total Group Employment/ Total CZ Population in 2005) × 100		
β: SC	−.262***	−.225***	−.204***
	(.079)	(.071)	(.074)
CZ-year trends	Yes	Yes	Yes
Bartiks	Yes	Yes	Yes
Y mean	3.05	2.52	2.26
% effect	−8.59	−8.95	−9.04
Observations	7,400	7,400	7,400
	B. Dependent Variable: Log Hourly Wages		
β: SC	−.014	−.003	−.012
	(.009)	(.016)	(.017)
CZ-year trends	Yes	Yes	Yes
Bartiks	Yes	Yes	Yes
Y mean	2.74	2.57	2.56
Observations	7,383	7,153	7,108

NOTE.—Data are drawn from the 2005–14 ACS. In panel A, the employment share is measured as total employment of the demographic group by CZ and year, divided by total working-age CZ population in 2005, and then multiplied by 100. In panel B, hourly wages are calculated by dividing annual earnings by annual hours worked among full-time workers. See sec. III for more details. Column 1 includes Hispanic foreign-born individuals with a high school degree or less. Column 2 further restricts the sample in col. 1 to include those who arrived in the United States after 1980. Column 3 includes working-age (ages 20–64) foreign-born individuals with a high school degree or less who were born in Mexico or Central America and arrived in the United States after 1980. The models include CZ fixed effects, year fixed effects, CZ-specific linear trends, and Bartik-style controls. The results are weighted using the CZ population in 2000. Standard errors (in parentheses) are clustered by CZ.

*** $p < .01$.

the wage results using these alternative definitions of likely undocumented immigrants in panel B are also similar in magnitude to the results on all LEFB individuals. We also show in table A5 that the estimated effects are similar when the sample of LEFB individuals is based on US citizenship (low-educated noncitizens) instead of country of birth (low-educated foreign-born individuals; Orrenius and Zavodny 2009; Amuedo-Dorantes and Bansak 2012).

We next check the robustness of the main results for LEFB and US-born workers to alternative samples and specifications, which we report in table 7 (employment share) and table 8 (hourly wages). For ease of comparison, we report the estimated effects in columns 1 and 4 from tables 3 and 4 in column 1 of tables 7 and 8. In column 2, we drop CZs that adopted SC before 2010, since these have been shown to be more selective on observable characteristics, such as proximity to the US-Mexico border (Cox and Miles 2013). We next drop CZs that adopted a sanctuary city policy before the implementation of

Table 7
Effects of SC on Employment Share of LEFB and US-Born Individuals: Robustness Tests

		Robustness to Dropping CZs			Control for Other Immigration Policies		Control for Economic Conditions	
	Baseline (1)	Drop Early Adoption (2)	Drop Sanctuary Cities (3)	Drop Arizona (4)	Control for 287(g) (5)	Control for E-Verify (6)	Region × Year Fixed Effects (7)	Quadratic Trend House Prices and Starts (8)
	A. LEFB Individuals							
β: SC	−.387***	−.258***	−.387***	−.315***	−.370***	−.370***	−.299***	−.390***
	(.093)	(.078)	(.093)	(.081)	(.108)	(.081)	(.097)	(.093)
CZ-year trends	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Bartiks	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Y mean	6.75	6.49	6.75	6.66	6.75	6.75	6.75	6.77
% effect	−5.74	−3.98	−5.74	−4.73	−5.49	−5.48	−4.42	−5.77
Observations	7,400	6,980	7,400	7,330	7,400	7,400	7,400	6,510
	B. US-Born Individuals							
β: SC	−.300*	−.377*	−.300*	−.343*	−.322*	−.285*	−.346*	−.300*
	(.168)	(.197)	(.168)	(.177)	(.176)	(.166)	(.191)	(.169)
CZ-year trends	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Bartiks	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Y mean	58.61	59.07	58.61	58.69	58.61	58.61	58.61	58.60
% effect	−.51	−.64	−.51	−.58	−.55	−.49	−.59	−.51
Observations	7,400	6,980	7,400	7,330	7,400	7,400	7,400	6,510

NOTE.—Data are drawn from the 2005–14 ACS. Panel A reports the effects on working-age foreign-born individuals with a high school degree or less. Panel B reports the effects on working-age US-born individuals. The employment share is measured as total employment in the demographic group by CZ and year, divided by total working-age CZ population in 2005, and then multiplied by 100. Column 2 drops CZs that adopted SC before 2010. Column 3 drops CZs that adopted sanctuary city policies before the adoption of SC. Column 4 drops CZs in Arizona. Columns 5 and 6 add controls for state and local 287(g) agreements and E-Verify policies, respectively. Column 7 adds controls for region × year fixed effects, using US Census divisions (see n. 23 for details). Finally, col. 8 adds quadratic trends multiplied by the preperiod change in housing prices and housing starts. The models include CZ fixed effects, year fixed effects, CZ-specific linear trends, and Bartik-style controls. The results are weighted using the CZ population in 2000. Standard errors (in parentheses) are clustered by CZ.

* $p < .10$.

*** $p < .01$.

Table 8
Effects of SC on Hourly Wage of LEFB and US-Born Individuals: Robustness Tests

		Robustness to Dropping CZs			Control for Other Immigration Policies		Control for Economic Conditions	
	Baseline (1)	Drop Early Adoption (2)	Drop Sanctuary Cities (3)	Drop Arizona (4)	Control for 287(g) (5)	Control for E-Verify (6)	Region × Year Fixed Effects (7)	Quadratic Trend House Prices and Starts (8)
	A. LEFB Individuals							
β: SC	−.0014	−.019*	−.014	−.015	−.017*	−.015*	−.021	−.015
	(.009)	(.010)	(.009)	(.009)	(.010)	(.009)	(.)	(.009)
CZ-year trends	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Bartiks	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Y mean	2.74	2.74	2.74	2.74	2.74	2.74	2.74	2.74
Observations	7,383	6,963	7,383	7,313	7,383	7,383	7,383	6,495
	B. US-Born Individuals							
β: SC	−.006**	−.006**	−.006**	−.007**	−.006**	−.006**	−.006**	−.006**
	(.003)	(.003)	(.003)	(.003)	(.002)	(.003)	(.003)	(.003)
CZ-year trends	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Bartiks	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Y mean	3.18	3.18	3.18	3.18	3.18	3.18	3.18	3.18
Observations	7,400	6,980	7,400	7,330	7,400	7,400	7,400	6,510

NOTE.—Data are drawn from the 2005–14 ACS. Panel A reports the effects on working-age foreign-born individuals with a high school degree or less. Panel B reports the effects on working-age US-born individuals. Hourly wages are calculated by dividing annual earnings by annual hours worked among full-time workers. Column 2 drops CZs that adopted SC before 2010. Column 3 drops CZs that adopted sanctuary city policies before the adoption of SC. Column 4 drops CZs in Arizona. Columns 5 and 6 add controls for state and local 287(g) agreements and E-Verify policies, respectively. Column 7 adds controls for region × year fixed effects, using US Census divisions (see n. 23 for details). Finally, col. 8 adds quadratic trends multiplied by the preperiod change in housing prices and housing starts. The models include CZ fixed effects, year fixed effects, CZ-specific linear trends, and Bartik-style controls. The results are weighted using the CZ population in 2000. Standard errors (in parentheses) are clustered by CZ.

* $p < .10$.
** $p < .05$.
*** $p < .01$.

SC in column 3, and in column 4, we drop Arizona, which passed measure SB 1070 in 2010, one of the broadest and most restrictive immigration enforcement measures in the country. Reassuringly, the results in columns 2–4 for both employment and wages are robust across all these different estimation samples for both LEFB and US-born workers.[22]

Our sample period overlaps with the rollout of both 287(g) agreements and E-Verify policies. We include these controls one at a time (287(g) in col. 5 and E-Verify in col. 6) to better understand how each program in isolation might affect our estimates, but the results are similar if we control for both at the same time. Controlling for the presence of either policy does not impact the estimated effects of SC. Thus, the results do not indicate that these enforcement policies confound the estimated effects of SC.

To further control for changes in local economic conditions, we include region × year fixed effects to account for the fact that different regions were impacted differently by the Great Recession.[23] The results in column 7 indicate that the addition of these fixed effects has little impact on the estimated effects of SC. Finally, we include quadratic trends in the 2000–2006 change in local housing prices and quadratic trends in the 2000–2006 change in the number of housing building permits issued in the CZ. These controls provide an alternative measure for the differential effects of the Great Recession. The results in column 8 do not meaningfully change from the baseline, providing further support that the implementation of SC was not correlated with the severity of the Great Recession.

D. Discussion of Results

We gauge the plausibility of the estimated effects of SC by first comparing them with historical policies that removed immigrants from the US labor market. Specifically, the fact that SC had no beneficial effects on US-born individuals' employment or wages is consistent with the findings of Clemens, Lewis, and Postel (2018), who showed that the removal of migrant labor through the ending of the Bracero program did not improve employment or wages of US native workers because firms adjusted their capital investments

[22] The results are also robust to dropping the largest 1% and 5% of CZs, which excludes the largest 80 and 370 CZs, respectively. This suggests that the effects of SC are not being identified off of a handful of large labor markets. These results are available upon request.

[23] Regions are defined on the basis of nine US Census divisions within the United States. These include New England, Middle Atlantic, East North Central, South Atlantic, East South Central, West North Central, West South Central, Mountain, and Pacific. Note that a few CZs span multiple divisions, and we assign them to the division to which the majority of their population belong. We do not include state × year fixed effects because 10 states and the District of Columbia implemented SC on a statewide basis. Moreover, many CZs span multiple states, making state × year fixed effects empirically difficult to implement.

and output goods. Lee, Peri, and Yasenov (2022) studied repatriations (forced out-migration) in the 1930s and estimated that a 1% decline in the population of Mexican migrants reduced the probability of being employed for incumbent native workers by 0.2%–0.3% and reduced overall native wages by 0.3%.[24] In comparison, our results imply that a 1% decrease in the employment share of LEFB individuals is associated with about a 0.1% decline in the employment share and wages of US-born individuals. We are unable to directly compare our results with the effects of 287(g) agreements because these papers do not estimate the effects separately for immigrants and natives (Pham and Van 2010; Bohn and Santillano 2017).[25]

Next, we compare our findings with previous studies on the impact of immigration inflows. We particularly focus on studies that use policy-driven variation in the inflow of immigrants across space and over time (Dustmann, Schönberg, and Stuhler 2016). Consistent with no overall benefit of removing immigrants on the outcomes of natives, Card (1990) and Peri and Yasenov (2019) found that a large influx of Cuban immigrants to the Miami labor market did not reduce the wages of low-educated native workers or increase their unemployment rates. More recently, Foged and Peri (2016) studied the impact of refugee inflows to Denmark on the labor market outcomes of Danish workers and found that a 1 percentage point increase in the employment share of low-skilled refugees led to a 0.8% increase in the fraction of the year worked by low-educated native Danish workers and a 1.8% increase in their hourly wage. This positive impact was driven by Danish workers who were able to move to less manual-intensive occupations, consistent with a complementarity in production between refugee and low-educated Danish workers. Our results (while of the opposite sign, since we look at immigrant outflows) are of a similar magnitude: a 1 percentage point decrease in the employment share of LEFB individuals led to a 1.3% decrease in the employment share of US-born workers (0.51/0.387) and a 1.6% decrease in their wages (0.6/0.387). An exception is the results of Dustmann, Schönberg, and Stuhler (2017), who found that an increase in commuter Czech workers to bordering German cities decreased the employment and wages of German workers. This may be because Czech workers did not live in Germany and so had a limited impact on local consumption, which, as

[24] Unlike in our analysis, Lee, Peri, and Yasenov (2022) found that US-born individuals moved within the United States in response to repatriations, so we focus on their estimates for individual workers rather than local labor markets, the latter of which is influenced by internal migration.

[25] A broad literature has also estimated the effect of an employment-based enforcement program on the labor outcomes of immigrants and natives. Research on E-Verify has found negative effects on labor outcomes of likely undocumented immigrants, with mixed evidence on the effects on documented immigrants and native workers (Amuedo-Dorantes and Bansak 2014; Orrenius and Zavodny 2015; Ayromloo, Feigenberg, and Lubotsky 2020).

detailed below, we find to be potentially important in explaining the negative effects of SC on US-born workers. Additionally, firms may have considered the commuting policy to be temporary, which prevented them from expanding their operations in response to the influx of workers.

VI. Sources of Changes in Labor Demand for US-Born Workers

In this section, we test for evidence of two possible mechanisms through which SC may have impacted the labor market outcomes of US-born workers: (1) changes in job competition and job creation and (2) changes in local consumption.

A. Job Competition and Job Creation

We first consider how SC may impact the demand for substitute and complement US-born workers. In the canonical model, immigrant and native workers are assumed to be perfect substitutes and earn the same wage, so the decline in the labor supply of undocumented immigrants is expected to increase the employment and wages of US-born workers. However, several studies have found that undocumented immigrants in the United States earn substantially lower wages compared with US-born workers. This could be due to differences in skill or to the fact that employers exert monopsony power over undocumented immigrants because of their legal status, which reduces their outside options, limits their access to the social safety net, and lowers their bargaining power (Rivera-Batiz 1999; Kossoudji and Cobb-Clark 2002; Pan 2012; Albert 2021). As a result, a reduction in the labor supply of undocumented immigrants is expected to increase the average labor costs that firms face, which decreases their ability to create new jobs. This decline in job creation counteracts the positive effects that US-born substitute workers experience from decreased job competition with undocumented immigrants. Therefore, as Chassamboulli and Peri (2015) and Albert (2021) show with a job search model, these two competing forces imply that the effect of a policy like SC on the labor market outcomes of substitute US-born workers is theoretically ambiguous.[26] In contrast, the decrease in job creation is expected to unambiguously decrease the demand for US-born workers who act as complements to low-educated workers (including LEFB individuals).

The predictions described above suggest that the spillover effects of SC on US-born workers should vary by their degree of substitution with undocumented immigrants. To proxy for this, as is common in much of the prior literature, we estimate the heterogeneous effects of SC by education, where

[26] The source of the decline in the competition faced by native workers differs between the two papers. While hiring is assumed to be random in Chassamboulli and Peri (2015), Albert (2021) assumes nonrandom hiring in which firms can distinguish between types of workers on the basis of their nativity and immigration status.

US-born workers with a high school degree or less are considered the closest substitutes for immigrant workers. The results in table 9 indicate that SC led to a statistically insignificant 0.093 percentage point decline in the employment share of low-educated US-born workers and a statistically significant 1.2% decline in their wages (col. 1). SC also reduced the employment share of US-born workers with more than a high school degree (high educated) by 0.207 percentage points (significant at the 10% level) and led to a statistically insignificant 0.4% decrease in their wages (col. 2). Thus, the evidence is consistent with a decline in the labor demand for both education groups of US-born workers possibly because of a decline in job creation.

Prior studies, however, have shown that immigrants downgrade their skills upon arrival and that native workers shift to less manually intensive occupations in response to migration (Dustmann, Frattini, and Preston 2012; Foged and Peri 2016). Thus, stratifying native workers only by education may be too coarse to allow us to identify who will be impacted by the

Table 9
Effects of SC on Employment Share and Log Hourly Wages of US-Born Individuals, by Education

	High School or Less (1)	More than High School (2)
	A. Dependent Variable: (Total Group Employment/ Total CZ Population in 2005) × 100	
β: SC	−.093	−.207*
	(.120)	(.115)
CZ-year trends	Yes	Yes
Bartiks	Yes	Yes
Y mean	23.59	35.02
% effect	−.39	−.59
Observations	7,400	7,400
	B. Dependent Variable: Log Hourly Wages	
β: SC	−.012***	−.004
	(.003)	(.003)
CZ-year trends	Yes	Yes
Bartiks	Yes	Yes
Y mean	2.94	3.31
Observations	7,400	7,400

NOTE.—Data are drawn from the 2005–14 ACS. The US-born sample includes working-age (ages 20–64) individuals. In panel A, the employment share is measured as total employment in the demographic group by CZ and year, divided by total working-age CZ population in 2005, and then multiplied by 100. In panel B, hourly wages are calculated by dividing annual earnings by annual hours worked among full-time workers. See sec. III for more details. All models include CZ and year fixed effects, CZ-specific linear trends, and Bartik-style controls. The results are weighted using the CZ population in 2000. Standard errors (in parentheses) are clustered by CZ.

* $p < .10$.
** $p < .05$.
*** $p < .01$.

job competition and job creation effects. We use an alternative method to estimate the heterogeneous effects of SC by stratifying the US-born workers into occupational skill categories (Peri and Sparber 2009; Ottaviano and Peri 2012). Specifically, we categorize occupations on the basis of the fraction of workers who have at least a college degree in each three-digit Standard Occupational Classification System occupation in 2005. We plot the distribution of occupational skills based on this measure in figure A3.[27] The median occupation has only about 13% of workers with a college degree, and the 25th and 75th percentiles have 5% and 42%, respectively. Thus, workers with at most a high school degree may still work in occupations with above-median occupational skill. On the other hand, workers with a college degree are unlikely to work in occupations at the bottom of the skill distribution.

We estimate heterogeneous effects on the employment share and wages of US-born workers within these occupational bins, separately for those with a high school degree or less and with more than a high school degree. We calculate the employment share by dividing the total number of workers in each occupational skill bin, education group, and CZ-year by the total number of working-age individuals in the CZ in 2005 and multiply by 100. We calculate wages by restricting the sample to the workers in the relevant occupational skill bin, education group, and CZ-year. For individuals who are not currently working, we use the occupation they report working in most recently.

To fix ideas about how job competition and job creation effects may differ by occupational skill, we plot the share of all LEFB workers in each occupational skill bin in 2005 (fig. 3*A*). We use a moving window approach in which we group occupations in the lowest quartile (0–25th) and then change this bin by five percentiles (e.g., 5th–30th, 10th–35th). As expected, the share of LEFB individuals is highest at the bottom of the distribution. These occupations include chefs and cooks, construction laborers, agricultural workers, and janitors and building cleaners. Figure 3*B* plots the estimated effects of SC on the employment share of low-educated US-born workers along the same occupational skill distribution. Interestingly, SC had a positive and marginally significant effect on the employment share of low-educated US-born workers at the bottom of the occupational skill distribution, where most undocumented immigrants work. These employment effects are consistent with the hypothesis that the benefits of reduced job competition for these workers outweigh the costs of reduced job creation. We find little evidence of changes in wages of low-educated US-born workers at the bottom of the occupational skill distribution (fig. 3*D*).

[27] The results are similar if we instead stratify occupations by average wages, or the percent of the occupation with less than a high school degree.

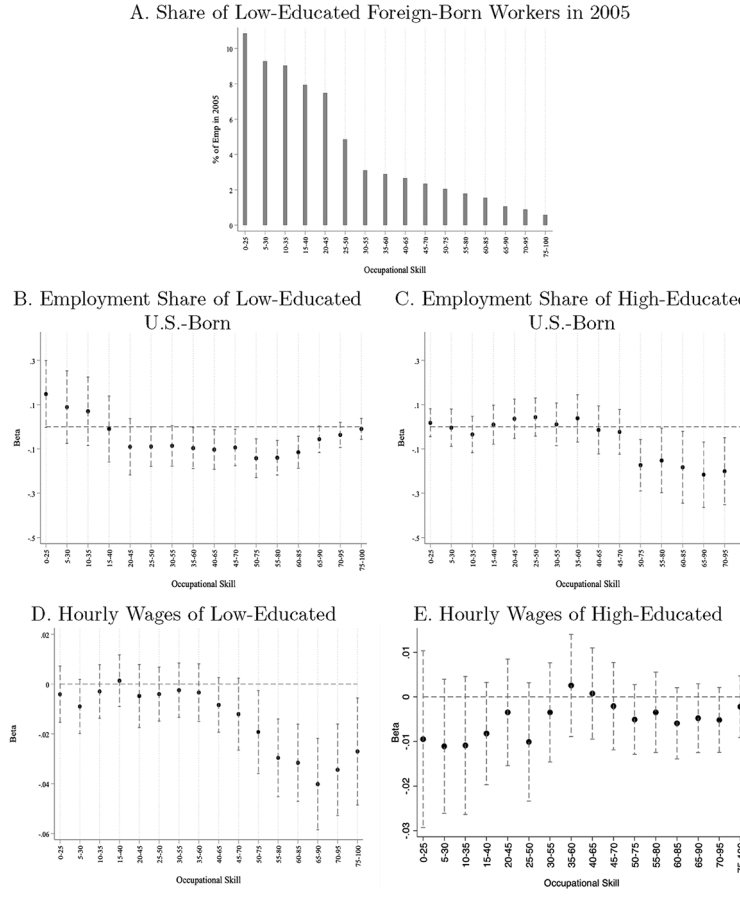


FIG. 3.—Effects of SC on employment share and wages of US-born individuals by occupational skill. Data are drawn from the 2005–14 ACS. The horizontal axis denotes occupational skill bins, starting from the lowest skill group (0–25th percentiles) on the left and increasing by five percentiles in each bin, moving to the right (e.g., 5th–30th, 10th–35th). *A* plots the share of all workers in 2005 in the occupational skill group of low-educated (high school degree or less) foreign-born individuals. *B* and *D* include a sample of low-educated (high school degree or less) US-born individuals aged 20–64. *C* and *E* include a sample of high-educated (more than a high school degree) US-born individuals aged 20–64. The employment share by occupational skill in *B* and *C* is calculated by dividing the number of workers in the demographic group, occupational skill group, CZ, and year by the total working-age CZ population in 2005 and then multiplying by 100. For individuals working, we use their current occupation, and for individuals not currently working, we use the most recent occupation they worked in. For wages, in *D* and *E*, we calculate the log of the hourly wages of workers in each occupational skill group. All models in *B–E* include CZ fixed effects, year fixed effects, CZ linear trends, and Bartik-style controls. These results are weighted using the CZ population in 2000, and standard errors are clustered by CZ. Dashed lines indicate 95% confidence intervals (*B–E*).

The effects of SC on the employment of low-educated US-born individuals become muted and then turn negative as we move up the occupational skill distribution. This is consistent with the idea that low-educated workers in higher-skilled occupations are less likely to benefit from the job competition effect, likely because they operate as complements in production, so the job creation mechanism dominates. This is supported by the fact that the occupations where we find the most negative effects on low-educated US-born workers can be described as supervisory of occupations with many undocumented immigrants, such as first-line supervisors of construction and production, construction managers, farmers and ranchers, and food service and lodging managers. Similarly, we find negative effects on low-educated US-born wages at the upper end of the occupational skill distribution. These findings are similar to those of Dustmann, Frattini, and Preston (2012), who showed that immigration inflows reduce the wages of native workers in the bottom 20th percentile of the wage distribution, where most migrant workers are located, while increasing the wages of natives in the upper part of the wage distribution.

Figure 3*C* plots the estimated effect on employment along the occupational skill distribution for US-born workers with more than a high-school degree. There is little evidence that SC impacted the employment share of high-educated US-born workers at the bottom of the occupational skill distribution, which likely reflects their low share of employment in these occupations. However, consistent with a decrease in job creation, SC lowered the employment share of high-educated workers in the upper half of the occupational skill distribution, who likely act as complements in production. There is little evidence of significant changes in wages for high-educated workers (fig. 3*E*).[28]

B. Local Consumption

The second mechanism through which SC can impact the labor market outcomes of US-born workers is reduced consumption for local goods. If this channel is an important driver of our overall results, we expect the negative effects of SC on US-born individuals to be larger in nontradable industries. Since tradable goods are consumed largely outside of the local labor market, the overall level of demand in these sectors—and therefore the prices of these goods—will be largely unaffected by local demand shocks. The demand for nontradable goods, on the other hand, is likely to go down in response to SC, which will impact prices and lead to a larger decline in these industries' demand for labor (Burstein et al. 2020).

[28] The results at the bottom of the occupational skill distribution for high-educated US-born individuals are noisy, likely because few high-educated individuals work in these occupations.

Table 10
Effects of SC on Employment Share and Log Hourly Wages of US-Born Individuals, by Tradability

	Employment Share (1)	Log Hourly Wages (2)
	A. Tradable	
β: SC	.128	−.009**
	(.097)	(.004)
CZ-year trends	Yes	Yes
Bartiks	Yes	Yes
Y mean	15.04	4.53
% effect	.85	. . .
Observations	7,400	7,400
	B. Nontradable	
β: SC	−.429***	−.006
	(.125)	(.004)
CZ-year trends	Yes	Yes
Bartiks	Yes	Yes
Y mean	43.57	4.53
% effect	−.98	. . .
Observations	7,400	7,400

NOTE.—Data are drawn from the 2005–14 ACS. The US-born sample includes working-age (ages 20–64) individuals. In col. 1, the employment share is measured as total employment in the demographic group by CZ and year, divided by total working-age CZ population in 2005, and then multiplied by 100. In col. 2, hourly wages are calculated by dividing annual earnings by annual hours worked among full-time workers. See sec. III for more details. We stratify the sample by whether the individual works in a tradable (panel A) or nontradable (panel B) industry. For individuals not currently working, we use the most recent industry they worked in. We classify industries as tradable or nontradable following Burstein et al. (2020). Tradable industries include agriculture and manufacturing, and nontradable sectors include construction, retail, wholesale, and nontradable services, such as haircuts. All models include CZ and year fixed effects, CZ-specific linear trends, and Bartik-style controls. The results are weighted using the CZ population in 2000. Standard errors (in parentheses) are clustered by CZ.

** $p < .05$.

*** $p < .01$.

To examine this mechanism, we analyze the effect of SC separately for tradable and nontradable industries and report these results in table 10.[29] Consistent with the hypothesis of a decline in local demand, we find a large, negative, and statically significant effect on employment of US-born workers within nontradable industries (col. 1 of panel B). The estimate indicates a 0.98% decline in the employment share, which is larger than the 0.51% decline across all industries (table 4). In the tradable sector, we find no evidence of an overall negative effect on employment (col. 1 of panel A). We find little difference in the effects of SC on the wages of US-born workers across tradable and nontradable industries (col. 2). Although these results

[29] We classify industries as tradable or nontradable, following Burstein et al. (2020). Tradable industries include agriculture and manufacturing, and nontradable sectors include construction, retail, wholesale, and nontradable services, such as haircuts.

suggest a potential role for the consumption channel, our reduced-form estimates cannot distinguish its relative importance compared with the job competition and job creation mechanism.

VII. Conclusion

SC, one of the largest federal immigration enforcement policies over the past decade, resulted in the deportation of almost half a million individuals during 2008–14. This paper makes an important contribution to the immigration literature by estimating the effects of this policy on the employment and wages of both likely undocumented immigrants and US-born individuals.

We find that SC led to a significant decrease in the employment share and wages of likely undocumented immigrants. Importantly, the results indicate that SC did not affect their labor market outcomes only through physical removals but also by reducing the employment and wages of those who remained in the United States.

The results further indicate that SC did not improve the overall employment or wages of US-born workers. In fact, we find that SC reduced the overall employment share and wages of US-born workers. We propose and test two potential mechanisms that may explain these results. First, any benefits from reduced job competition that US-born workers face may be counteracted by a decline in labor demand due to an increase in labor costs when there are fewer undocumented immigrants in the labor force. Second, the reduction in the population and earnings of likely undocumented immigrants may reduce local consumption, which would also reduce labor demand for US-born workers. Our results suggest that both mechanisms may be potentially important.

On the basis of our findings, police-based enforcement policies aimed at reducing the number of undocumented immigrants should consider the potential negative spillover effects on the labor market outcomes of immigrants who remain in the United States and on US-born workers. The fiscal costs of operating SC, including the costs of detentions, were roughly $1 billion per year (*Houston Chronicle* 2008), and our results provide important evidence about additional costs due to these negative spillover effects on US-born workers.

References

Albert, Christoph. 2021. "The Labor Market Impact of Immigration: Job Creation versus Job Competition." *American Economic Journal: Macroeconomics* 13, no. 1:35–78.

Alsan, Marcella, and Crystal Yang. 2018. "Fear and the Safety Net: Evidence from Secure Communities." NBER Working Paper no. 24731, National Bureau of Economic Research, Cambridge, MA.

Altonji, Joseph G., and David Card. 1991. "The Effects of Immigration on the Labor Market Outcomes of Less-Skilled Natives." In *Immigration, Trade, and the Labor Market*, ed. John M. Abowd and Richard B. Freeman, 201–34. Chicago: University of Chicago Press.

Amuedo-Dorantes, Catalina, and Cynthia Bansak. 2012. "The Labor Market Impact of Mandated Employment Verification Systems." *American Economic Review* 102, no. 3:543–48.

———. 2014. "Employment Verification Mandates and the Labor Market Outcomes of Likely Unauthorized and Native Workers." *Contemporary Economic Policy* 32, no. 3:671–80.

Amuedo-Dorantes, Catalina, Thitima Puttitanun, and Ana P. Martinez-Donate. 2019. "Deporting 'Bad Hombres'? The Profile of Deportees under Widespread versus Prioritized Enforcement." *International Migration Review* 53, no. 2:518–47.

Anelli, Massimo, Gaetano Basso, Giuseppe Ippedico, and Giovanni Peri. 2023. "Emigration and Entrepreneurial Drain." *American Economic Journal: Applied Economics* 15, no. 2:218–52.

Autor, David H., and David Dorn. 2013. "The Growth of Low-Skill Service Jobs and the Polarization of the US Labor Market." *American Economic Review* 103, no. 5:1553–97.

Autor, David H., David Dorn, and Gordon H. Hanson. 2013. "The China Syndrome: Local Labor Market Effects of Import Competition in the United States." *American Economic Review* 103, no. 6:2121–68.

Ayromloo, Shalise, Benjamin Feigenberg, and Darren Lubotsky. 2020. "States Taking the Reins? Employment Verification Requirements and Local Labor Market Outcomes." NBER Working Paper no. 26676, National Bureau of Economic Research, Cambridge, MA.

Bansak, Cynthia, and Sarah Pearlman. 2021. "Endogamous Marriage among Immigrant Groups: The Impact of Deportations under Secure Communities." GLO Discussion Paper no. 756, Global Labor Organization, Essen.

Bartik, Timothy. 1992. "The Effects of State and Local Taxes on Economic Development: A Review of Recent Research." *Economic Development Quarterly* 6, no. 1:102–10.

Bodvarsson, Orn B., Joshua J. Lewer, and Hendrik F. Van den Berg. 2008. "Measuring Immigration's Effects on Labor Demand: A Reexamination of the Mariel Boatlift." *Labour Economics* 15, no. 4:560–74.

Bohn, Sarah, and Robert Santillano. 2017. "Local Immigration Enforcement and Local Economies." *Industrial Relations: A Journal of Economy and Society* 56, no. 2:236–62.

Borjas, George J. 2003. "The Labor Demand Curve Is Downward Sloping: Reexamining the Impact of Immigration on the Labor Market." *Quarterly Journal of Economics* 118, no. 4:1335–74.

Borjas, George J., and Hugh Cassidy. 2019. "The Wage Penalty to Undocumented Immigration." *Labour Economics* 61:101757.

Borjas, George J., and Lawrence F. Katz. 2007. "The Evolution of the Mexican-Born Workforce in the United States." In *Mexican Immigration to the United States*, ed. George Borjas, 13–56. Chicago: University of Chicago Press.

Borusyak, Kirill, Xavier Jaravel, and Jann Spiess. 2022. "Revisiting Event Study Designs: Robust and Efficient Estimation." Working paper.

Boustan, Leah Platt, Price V. Fishback, and Shawn Kantor. 2010. "The Effect of Internal Migration on Local Labor Markets: American Cities during the Great Depression." *Journal of Labor Economics* 28, no. 4:719–46.

Brown, J. David, Misty L. Heggeness, Suzanne M. Dorinski, Lawrence Warren, and Moises Yi. 2018. "Understanding the Quality of Alternative Citizenship Data Sources for the 2020 Census." Working Paper no. 18-38, Center for Economic Studies, US Census Bureau, Washington, DC.

Burstein, Ariel, Gordon Hanson, Lin Tian, and Jonathan Vogel. 2020. "Tradability and the Labor-Market Impact of Immigration: Theory and Evidence from the United States." *Econometrica* 88, no. 3:1071–112.

Cadena, Brian C., and Brian K. Kovak. 2016. "Immigrants Equilibrate Local Labor Markets: Evidence from the Great Recession." *American Economic Journal: Applied Economics* 8, no. 1:257–90.

Callaway, Brantly, and Pedro H. C. Sant'Anna. 2021. "Difference-in-Differences with Multiple Time Periods." *Journal of Econometrics* 225, no. 2:200–230.

Card, David. 1990. "The Impact of the Mariel Boatlift on the Miami Labor Market." *Industrial and Labor Relations Review* 43, no. 2:245–57.

Chassamboulli, Andri, and Giovanni Peri. 2015. "The Labor Market Effects of Reducing the Number of Illegal Immigrants." *Review of Economic Dynamics* 18, no. 4:792–821.

Clemens, Michael A., Ethan G. Lewis, and Hannah M. Postel. 2018. "Immigration Restrictions as Active Labor Market Policy: Evidence from the Mexican Bracero Exclusion." *American Economic Review* 108, no. 6: 1468–87.

Cox, Adam B., and Thomas J. Miles. 2013. "Policing Immigration." *University of Chicago Law Review* 80, no. 1:87–136.

Dorn, David. 2009. "Essays on Inequality, Spatial Interaction, and the Demand for Skills." PhD diss., University of St. Gallen.

Dustmann, Christian, Francesca Fabbri, and Ian Preston. 2005. "The Impact of Immigration on the British Labour Market." *Economic Journal* 115, no. 507:F324–F341.

Dustmann, Christian, Tommaso Frattini, and Ian P. Preston. 2012. "The Effect of Immigration along the Distribution of Wages." *Review of Economic Studies* 80, no. 1:145–73.

Dustmann, Christian, Tommaso Frattini, and Anna Rosso. 2015. "The Effect of Emigration from Poland on Polish Wages." *Scandinavian Journal of Economics* 117, no. 2:522–64.

Dustmann, Chtistian, Uta Schönberg, and Jan Stuhler. 2016. "The Impact of Immigration: Why Do Studies Reach Such Different Results?" *Journal of Economic Perspectives* 30, no. 4:31–56.

———. 2017. "Labor Supply Shocks, Native Wages, and the Adjustment of Local Employment." *Quarterly Journal of Economics* 132:435–83.

East, Chloe N., and Andrea Velásquez. 2022. "Unintended Consequences of Immigration Enforcement: Household Services and Highly-Educated Females' Work." *Journal of Human Resources* (May): 0920-11197R1.

Elsner, Benjamin. 2013. "Emigration and Wages: The EU Enlargement Experiment." *Journal of International Economics* 91, no. 1:154–163.

Foged, Mette, and Giovanni Peri. 2016. "Immigrants' Effect on Native Workers: New Analysis on Longitudinal Data." *American Economic Journal: Applied Economics* 8, no. 2:1–34.

Fuest, Clemens, Andreas Peichl, and Sebastian Siegloch. 2018. "Do Higher Corporate Taxes Reduce Wages? Micro Evidence from Germany." *American Economic Review* 108, no. 2:393–418.

Genoni, Maria, Gabriela Farfan, Luis Rubalcava, Graciela Teruel, Duncan Thomas, and Andrea Velasquez. 2017. "Mexicans in America." BREAD Working Paper no. 511, Bureau for Research and Economic Analysis of Development.

Goodman-Bacon, Andrew. 2021a. "Difference-in-Differences with Variation in Treatment Timing." *Journal of Econometrics* 225, no. 2:254–77.

———. 2021b. "The Long-Run Effects of Childhood Insurance Coverage: Medicaid Implementation, Adult Health, and Labor Market Outcomes." *American Economic Review* 111, no. 8:2550–93.

Greenwood, Michael J., and Gary L. Hunt. 1984. "Migration and Interregional Employment Redistribution in the United States." *American Economic Review* 74, no. 5:957–69.

Hercowitz, Zvi, and Eran Yashiv. 2002. "A Macroeconomic Experiment in Mass Immigration." IZA Discussion Paper no. 475, Institute of Labor Economics.

Hines, Annie, and Giovanni Peri. 2019. "Immigrants' Deportations, Local Crime and Police Effectiveness." IZA Discussion Paper no. 12413, Institute of Labor Economics, Bonn.

Hoefer, Michael, Nancy Francis Rytina, and Brian Baker. 2012. *Estimates of the Unauthorized Immigrant Population Residing in the United States: January 2011*. Department of Homeland Security, Office of Immigration Statistics. http://www.dhs.gov/xlibrary/assets/statistics/publications/ois_ill_pe_2011.pdf.

Houston Chronicle. 2008. "Possible Solutions for Immigrant Inmate Screenings." *Houston Chronicle*, November 16. https://www.chron.com/news/article/Possible-solutions-for-immigrant-inmate-screenings-1784152.php.

Kohli, Aarti, Peter L. Markowitz, and Lisa Chavez. 2011. "Secure Communities by the Numbers: An Analysis of Demographics and Due Process." Berkeley: Warren Institute on Law and Social Policy.

Kossoudji, Sherrie A., and Deborah Cobb-Clark. 2002. "Coming out of the Shadows: Learning about Legal Status and Wages from the Legalized Population." *Journal of Labor Economics* 20, no. 3:598–628.

Lee, Jin Young, and Gary Solon. 2011. "The Fragility of Estimated Effects of Unilateral Divorce Laws on Divorce Rates." *BE Journal of Economic Analysis and Policy* 11, no. 1:49.

Lee, Jongkwan, Giovanni Peri, and Vasil Yasenov. 2022. "The Labor Market Effects of Mexican Repatriations: Longitudinal Evidence from the 1930s." *Journal of Public Economics* 205:104558.

Miles, Thomas J., and Adam B. Cox. 2014. "Does Immigration Enforcement Reduce Crime? Evidence from Secure Communities." *Journal of Law and Economics* 57, no. 4:937–73.

Mishra, Prachi. 2007. "Emigration and Wages in Source Countries: Evidence from Mexico." *Journal of Development Economics* 82, no. 1:180–99.

Orrenius, Pia M., and Madeline Zavodny. 2009. "The Effects of Tougher Enforcement on the Job Prospects of Recent Latin American Immigrants." *Journal of Policy Analysis and Management* 28, no. 2:239–57.

———. 2015. "The Impact of E-Verify Mandates on Labor Market Outcomes." *Southern Economic Journal* 81, no. 4:947–59.

Ottaviano, Gianmarco, and Giovanni Peri. 2012. "Rethinking the Effect of Immigration on Wages." *Journal of the European Economic Association* 10, no. 1:152–97.

Pan, Ying. 2012. "The Impact of Legal Status on Immigrants' Earnings and Human Capital: Evidence from the IRCA 1986." *Journal of Labor Research* 33:119–42.

Passel, Jeffrey S., and D'Vera Cohn. 2011. "Unauthorized Immigrant Population: National and State Trends, 2010." Washington, DC: Pew Research Center.

———. 2014. "Unauthorized Immigrant Totals Rise in 7 States, Fall in 14." Washington, DC: Pew Research Center.

———. 2016a. "Overall Number of U.S. Unauthorized Immigrants Holds Steady since 2009." Washington, DC: Pew Research Center.

———. 2016b. "Size of U.S. Unauthorized Immigrant Workforce Stable after the Great Recession." Washington, DC: Pew Research Center.

Peri, Giovanni, and Chad Sparber. 2009. "Task Specialization, Immigration, and Wages." *American Economic Journal: Applied Economics* 1, no. 3:135–69.

Peri, Giovanni, and Vasil Yasenov. 2019. "The Labor Market Effects of a Refugee Wave: Synthetic Control Method Meets the Mariel Boatlift." *Journal of Human Resources* 54, no. 2:267–309.

Pham, Huyen, and Pham Hoang Van. 2010. "The Economic Impact of Local Immigration Regulation: An Empirical Analysis." *Cardozo Law Review* 32:485–518.

Rivera-Batiz, Francisco L. 1999. "Undocumented Workers in the Labor Market: An Analysis of the Earnings of Legal and Illegal Mexican Immigrants in the United States." *Journal of Population Economics* 12, no. 1:91–116.

Ruggles, Steven, Katie Genadek, Ronald Goeken, Josiah Grover, and Matthew Sobek. 2017. "Integrated Public Use Microdata Series: Version 7.0." Minneapolis: University of Minnesota.

Schmidheiny, Kurt, and Sebastian Siegloch. 2020. "On Event Studies and Distributed-Lags in Two-Way Fixed Effects Models: Identification, Equivalence, and Generalization." CEPR Discussion Paper no. 13477, Center for Economic Policy Research, London.

Suarez Serrato, Juan Carlos, and Owen Zidar. 2016. "Who Benefits from State Corporate Tax Cuts? A Local Labor Markets Approach with Heterogeneous Firms." *American Economic Review* 106, no. 9:2582–624.

Tolbert, Charles M., and Molly Sizer. 1996. "U.S. Commuting Zones and Labor Market Areas: A 1990 Update." ERS Staff Paper no. 9614, Economic Research Service, Washington, DC.

US Bureau of Labor Statistics. 2021. "Labor Force Statistics from the Current Population Survey." Washington, DC: US Bureau of Labor Statistics. https://www.bls.gov/cps/definitions.htm#:~:text=In%20other%20words%2C%20it%.

Valdivia, Carolina. 2019. "Expanding Geographies of Deportability: How Immigration Enforcement at the Local Level Affects Undocumented and Mixed-Status Families." *Law and Policy* 41, no. 1:103–19.

Van Hook, Jennifer, and James Bachmeier. 2013. "Citizenship Reporting in the American Community Survey." *Demographic Research* 29, no. 1:1–32.

Van Hook, Jennifer, Frank D. Bean, James D. Bachmeier, and Catherine Tucker. 2014. "Recent Trends in Coverage of the Mexican-Born Population of the United States: Results from Applying Multiple Methods across Time." *Demography* 51, no. 2:699–726.

Wang, Julia Shu-Huah, and Neeraj Kaushal. 2018. "Health and Mental Health Effects of Local Immigration Enforcement." *International Migration Review* 53, no. 4:970–1001.

Warren, Robert. 2014. "Democratizing Data about Unauthorized Residents in the United States: Estimates and Public-Use Data, 2010 to 2013." *Journal on Migration and Human Security* 2, no. 4:305–28.

Warren, Robert, and John Robert Warren. 2013. "Unauthorized Immigration to the United States: Annual Estimates and Components of Change, by State, 1990 to 2010." *International Migration Review* 47, no. 2:296–329.

Watson, Tara. 2013. "Enforcement and Immigrant Location Choice." NBER Working Paper no. 19626, National Bureau of Economic Research, Cambridge, MA.

Wolfers, Justin. 2006. "Did Unilateral Divorce Laws Raise Divorce Rates? A Reconciliation and New Results." *American Economic Review* 96, no. 5:1802–20.